

Architectural Particular Specification

SPECIAL COATINGS

SECTION 8.9

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SECTION 8.9

SPECIAL COATINGS

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The Work of this Section includes, but is not limited to, the furnishing and installing of the following as indicated in drawings:
 - 1. Surface preparation and field application of special coatings in designated areas.
 - 2. Substrate and material compatibility testing.
 - 3. Protection of adjacent and adjoining surfaces.
 - 4. Carpark deck coating system.

1.02 REFERENCES

- A. Standards
 - 1. BS 6151 : 1992 - Code of Practice for Painting of Buildings
 - 2. BS 3900 : All Parts - Method of Test for Paints (date varies) of Determination of Volatile and Non-Volatile Content
 - 3. BS 6782: Part 1: 1982 (1921) - Binders for Paint: Method of Determination of Volatile and Non-Volatile Content.
 - 4. Or other Applicable Standards.

1.03 QUALITY ASSURANCE

- A. Qualification of Special Coating Installer/Applicator:
 - 1. Company specializing in the installation of the systems specified herein, with minimum of five (5) years documented experience, certified and approved by the coating manufacturers specified herein.
- B. Qualification of Paint Coating Manufacturers:
 - 1. Specializing in the manufacturing of high performance, commercial grade paint coating systems and materials for at least ten (10) years.
 - 2. The paint coating Manufacturers shall employ a Field Technical Representative available for initial, interim and final Project inspections. The Manufacturer's Field Technical Representative shall suitable experience and knowledge of the coating material and its application on projects of similar size and complexity.
 - 3. The paint coating manufacturer shall be capable of providing field service representation during construction, approving acceptable installers, application method, and conducting final inspection of the paint coating system/assembly.
- C. A detailed QA/QC system is to be provided by the Contractor for approval for the epoxy system and to form part of the overall project QA/QC system.
- D. The QA/QC system is to provide for random testing on abrasion and pull off test of the Epoxy at a rate of 2 samples per 300m² or part thereof. The testing is to be carried out by an approved independent laboratory.
- E. A moisture test is required to carry out prior to the installation of epoxy system. Inform the Architect (with 24 hours notice) that manufacturer's tests are to be carried out to prove substrates are ready to receive epoxy, to permit test witnessing. Submit manufacturer's application manual, which describes the preparation of surfaces and application of specified materials.

1.04 PERFORMANCE REQUIREMENTS

- A. All material, special coating products and accessories shall be new, unused and not expired.

1.05 ENVIRONMENTAL REQUIREMENTS

- A. All primers, special coatings, solvents and other accessory materials shall be lead free and low in volatile organic compounds (VOC) in strict conformance to BS7557, BS6782: Part 1, and ASTM D3960.
- B. Apply special coatings only when moisture content and PH value of surfaces is within acceptable ranges recommended by the Manufacturers.
- C. Provide adequate continuous ventilation and sufficient heating facilities to maintain proper temperatures before, during, and 48 hours after application of coating as recommended.

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

- A. The Contractor and the Manufacturers are deemed to be familiar with psychometric perimeters used by the Consulting Mechanical Engineer in determining indoor temperature and relative humidity requirements. Indoor air quality requirements and purging procedures.

1.07 SUBMITTALS

- A. Submittals as indicated in Section 1.5.
- B. 300mm x 300mm samples of all specified colors and coating systems. Submit 3 different colours, based on architects sample, of varying hue and intensity of colour, as directed by the architect.
- C. Material hazard, health, safety and handling instruction.

Test Reports and Manufacturers' Product Data on all special coatings, primers, solvents, special coating products, and accessories.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY, STORAGE, AND HANDLING

- A. Deliver paint materials in sealed original labelled containers, bearing the Manufacturer's name, type of paint, brand name, color designation, and instructions for mixing and/or reducing, pot life/expiration date, and lot number.
- B. Provide adequate storage facilities. Store special coating materials at minimum ambient temperature recommended by the Manufacturers in well ventilated area.
- C. Take precautionary measures to prevent fire hazards and spontaneous combustion.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

(NOT USED)

1.12 WARRANTY AND MAINTENANCE BOND

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

- B. The Contractor and Special Coatings Manufacturers shall warrant that all products, material and items are suitable, and can be maintained in the various climatic and psychometric conditions encountered without adhesion or cohesion failures, blistering, chalking, fading, discoloration, organic and microbial (fungal or algal) growth and other defects for a period of ten (10) years after the practical completion of the building.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS/ SUPPLIERS

A. Subject to compliance with the Project Contract Documents, provide special coatings and required accessories and materials from one of the following manufacturers/ suppliers, or equivalent approved by the Architect:

1. REMMERS Viscacid Heavy Duty Epoxy Floor Screed System for Basement Carpark and G/F Loading/Unloading area, its components including:

Primer: REMMERS Epoxy ST 100 (Formerly known as REMMERS Viscacid Construction Resin 50)

Body Coat: REMMERS Epoxy OS Color New (Formerly known as REMMERS Viscacid Epoxy Coating OS) and silica sand

For Indoor Top Coat: REMMERS Epoxy OS Color New (Formerly known as REMMERS Viscacid Epoxy Coating OS)

For Outdoor Top Coat: REMMERS PUR Top OS 11

Tech Link Construction Engineering Limited

Unit 802, Lafort Centre,

838 Lai Chi Kok Road,

Kowloon, Hong Kong

Tel: (852) 3701 1817 / 9316 8906 Fax: (852) 3701 1818

Email: kenuskwong@techlinkcon.com.hk

2. The REMMERS Light Duty Epoxy Anti-slip Coating System for wall and floor of plant room/pump room, its components including:

Primer: REMMERS Viscacid BS2000, medium solids, environmentally-friendly, low odour, water emulsifiable, water vapor permeable, Coloured 2-component epoxy resin coating sprinkled with silica sand;

Top Coat: REMMERS Viscacid BS3000, High solids, low odour, environmentally correct, abrasion resistant, water emulsifiable, water vapor permeable, coloured 2-component epoxy coating in matt finish/silk gloss finish.

Tech Link Construction Engineering Limited

Unit 802, Lafort Centre,

838 Lai Chi Kok Road,

Kowloon, Hong Kong

Tel: (852) 3701 1817 / 9316 8906 Fax: (852) 3701 1818

Email: kenuskwong@techlinkcon.com.hk

3. The BASF Mastertop 1705

Tech Link Construction Engineering Limited

Unit 802, Lafort Centre,

838 Lai Chi Kok Road,

Kowloon, Hong Kong

Tel: (852) 3701 1817 / 9316 8906 Fax: (852) 3701 1818

Email: kenuskwong@techlinkcon.com.hk

2.02 MATERIALS

A. REMMERS Viscaid Heavy Duty Epoxy Floor Screed System for Basement Carpark and G/F Loading / Unloading Area, its minimum performance shall be as follows:

1. Every layer of the system must be from the same source of manufacturer and compatible with each other;
2. The manufacturer must have a minimum of 20 years' experience in related business;
3. The manufacturer must have the quality standards of EN ISO 9001 and EN ISO 14001;
4. Valid Agency Agreement for the distributorship of the Supplier in Hong Kong from the manufacturer must be submitted for approval;
5. The manufacturer shall supply a complete VOC certified product range which meets the strict criteria of the Swiss AgBB;
6. Other properties are shown as below

a. Minimum Performance of "REMMERS Resin 50"

1) Proprietary Epoxy Construction Resin

- Epoxy Construction Resin can be used for the production of compression resistant mortars, flow mortars and self-flowing thick coatings. The hardened coverings are resistant to water and aqueous chemical solutions and are not attacked by petrol, fats and oils. The material hardens shrink-free and can withstand heavy mechanical loads.
- It can also be used as a trowellable epoxy screed on mechanically heavy loadable industrial floor surfaces, as a repair and bedding mortar for concrete roadways and other load-bearing building elements. Besides, it can be used as a binder for producing fine or coarse anti-slip coverings and blinded finish floor coverings, as well as a binder for mortars and bonding layers. Other properties are shown as below:

Properties	Test Method	Result
Shore hardness	DIN 53505 D-Shore	80
Optical surface evaluation	ISO 4618-2 ISO 4628-1	Blistering 0 Streaks 0 Runs 0 Lifting 0 Frosted 0 Holes 0 Orange skin effect 0 Needle holes 0 Visible seams 0 Bleeding 0 Contraction 0 White chilling 0 Crazeing 0 Cell formation 0 0 = no perceptible change
Early water resistance	DIN 50014-23/50-2	Chilling could not be discerned
Static E-modulus	1. DIN 53457	Epoxy Construction Resin

	2. DIN 1048: Part 5	50: 1200 N/mm ² Epoxy Construction Resin 50 Mortar 1:9 silica sand: 27000 N/mm ²
Flexural / tensile strength	DIN EN 196 T1	Epoxy Construction Resin 50: 77.7 N/mm ² Epoxy Construction Resin 50 Mortar 1:9 silica sand: 34.9 N/mm ²
Compressive strength	DIN EN 196 T1	Epoxy Construction Resin 50: 74.5 N/mm ² Epoxy Construction Resin 50 Mortar 1:9 silica sand: 101.7 N/mm ²
Viscosity (flow time)	DIN 53211	107 mean values(s)
Thermal expansion coefficient	ASTM 3386-84	-20°C / +23°C : 17.7 x 10-6K-1 +23°C / +40°C : 23.8 x 10-6K-1 -20°C / +40°C : 19.4 x 10-6K-1

b. Minimum Performance of REMMERS Resin 50 + OS :

1) Proprietary Epoxy Construction Resin + Epoxy Coating OS

Properties	Test Method	Result
Linear shrinkage	DIN EN 12317-1	1. Degree of shrinkage S1 after 7 days storage at (21±1)°C and (60±10)% relative humidity : 0.11% 2. Degree of shrinkage S2 after 1 day storage at (70±2)°C and 1 day storage at (21±2)°C : 0.24%
Compressive strength	DIN EN 12190	81.7 N/mm ²
Co2 permeability	DIN EN 1062-6	CO2 diffusion equivalent air layer thickness SD CO2 diffusion resistance coefficient μ > 200m > 40,000
Water Vapour Permeability System construction I System construction II	DIN EN 7783-1	Water vapour diffusion equivalent air layer current density V thickness SD 119m 0.17 g/m ² d 482m 0.05 g/m ² d
Capillary water absorption and water permeability	DIN EN 1062-3	0.005 kg/m ² h0.5
Adhesive strength after test of	DIN EN 13687-3	

compatibility of alternating temperature Alternating frost-thaw load without de-icing salt attack (20 x) System construction I System construction II		2.9 N/mm ² 2.5 N/mm ²
Adhesive strength after test of compatibility with alternating temperature – for applications in outdoor areas influenced by de-icing salts (For applications in outdoor areas under influence of de-icing salts) Thunder shower load (Temperature shock) (10x) and frost-thaw load with de-icing salt attack (50x) System construction I System construction II	DIN EN 13687-2 DIN EN 13687-1	2.6 N/mm ² 3.1 N/mm ²
Resistant to strong Chemical Attack Test liquids: groups 1, 3 and 10 Petrol Heating oil, diesel fuel and unused motor and gear oils 10. Inorganic acids up to 20% and acid based, hydrolysing salts in aqueous solutions (pH < 6) with the exception of hydrofluoric acid and oxidising acids and their salts	DIN EN 13529	Reduction of Shore hardness 24 h after end of contact with medium Test liquid 1: -6 % Test liquid 3: ± 0 % Test liquid 10: -5 %

Impact resistance	DIN ISO 6272-2	No cracks or detachment
Pull off test System construction I	DIN EN 1542 Test specimen produced at: 1. 20°C 2. 8°C	1. 3.7 N/mm ² 2. 3.9 N/mm ²
Behaviour under fire after application	DIN EN 13501-1	Class Efl
Skid resistance / slip resistance System construction I	DIN EN 13036-4	60 SRT units

c. Minimum Performance of “REMMERS PUR Top OS 11”

Properties	Test Result
Reaction to fire	Bfl ³⁾
Release of corrosive substances	SR
Water permeability	NPD ²⁾
Wear resistance	AR 1 ¹⁾
Adhesive pull strength	B 1.5
Impact resistance	IR 4
Sound insulation	NPD
Sound absorption	NPD

1) Determined according to the BCA method on smooth covers.

2) NPD: No Performance Determined

3) In Germany DIN 4102 still applies, fire rating B1 is fulfilled and comparable with DIN EN 13501-1 Class B1.

d. Minimum performance of REMMERS Epoxy BS 2000 New:

REMMERS Viscacid BS 2000 (Epoxy BS2000) medium solids, water emulsifiable, water vapor permeable, low odour, durable, 2-component epoxy coating primer is widely used as a 2-component 100% reactive multi functional epoxy resin primer that can be diluted with water for the purpose of impregnation into the substrate. Once cured, it must prevent the absorption of noxious substances dissolved in water and increased resistance to wear and tear. It shall have the following additional properties:

a) Properties	Value	Test Method
Density	1.38-1.40g/cm ³	DIN 1045 /1048
Solid content	55% by weight	DIN1045/1048
Viscosity	800 m Pa.S	
Adhesion Strength	3.90N/mm ² (concrete failure)	EN1348:97
Abrasion Resistance	max0.25 gm	CS 17
Odour	None	
Toxicity	Physically Safe	
Thermal Resistance	0-80 degree C	
Loading Capacity	24 hours (Foot Traffic), 7 days (Full Strength)	

b) Table 1 : The Formaldehyde results

Test	Result	Test Method
Formaldehyde Content	Not Detected ^a	The sample was analysed by UV-Vis Spectrophotometer using Acetylacetone as reagent.

^a – The method detection limit was 0.1%

c) Table 2 : The Elemental results

Test	Result	Test Method
Mercury	Not Detected ^b	The sample was digested in Inorganic acid, followed by analysis using Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP-AES)
Lead	Not Detected ^b	
Cadmium	Not Detected ^b	
Chromium	Not Detected ^b	

^b – The method detection limit was 0.1%

d) Table 3 : The Flash Point results

Test	Result	Test Method
Flash point @61°C	No Flash	By Seta Flash

e) Table 4 : The Volatile Organic Compound (VOC) content

Test	Result	Test Method
VOC content ^c	41.8 g/L	By BS EN ISO 11890-2:2006, Paints and varnishes – Determination of content volatile organic compound (VOC) content – Part 2: Gas-chromatographic method.

^c – Volatile organic compound (VOC) means any organic compound having an initial boiling point less than or equal to 250°C measured at a standard pressure of 101,3kPa.

f) Table 5 : The analytical results

Test	Result	Test Method
Halogenated solvent	Not Detected ^d	By Gas Chromatograph – Mass Spectrometry (GC-MS)
Epichlorohydrin	Not Detected ^d	
Aromatic solvent	Not Detected ^e	

^d – The method detection limit was 0.01%

^e – The method detection limit was 0.1%

e. Minimum performance of REMMERS Epoxy BS 3000 New:

REMMERS Viscacid BS 3000 (Epoxy BS3000) is an environmental-friendly, high solids, low odour, water emulsifiable, water vapor permeable, coloured 2-component epoxy resin coating in matt or silk gloss finish. When applied as coating, it will not result in the encapsulation of vapor or moisture in concrete. It has resistant to petrol, thinned acids and alkalines. It shall have the following additional properties:

a) Adhesive pull strength

on concrete:	> 3.7 N/mm ² concrete failure
on old epoxy coat:	> 3.1 N/mm ² concrete failure
on old polyurethane coat	> 2.7 N/mm ² concrete failure

b) Abrasion according to Taber (1000 g load / 1000 rev.)

Test roll CS 17: < 0.10g
Test roll CS 10: < 0.06g

c) Water vapour diffusion as a coating:

Sd value < 2m

d) Hardening time at 20°C / 60% relative air humidity:

Dust dry after:	approx. 2 hours
Foot traffic after:	approx. 8 hours
Can be worked over:	approx. 8-48 hours
Thoroughly hardened:	approx. 7 days
Rain tight:	approx. 8 hours

e) Pendulum hardness according to König: 92 sec.

f) Available Colors are Ca. RAL 6005 Moosgrün, Ca. RAL 1001 Beige, Ca. RAL 7012 Basaltgrau, Ca. RAL 8012 Rotbraun, Ca. RAL 7035 Lichtgrau, Ca. RAL 1013 Perlweiß, Ca. RAL 1023 Verkehrsgelb, Ca. RAL 7001 Silbergrau, Ca. RAL 7032 Kieselgrau, Ca. RAL 6011 Resedarün, Ca. RAL 3011 Braunrot, Ca. RAL 7023 Betongrau and Ca. RAL 5014 Taubenblau.

g) Table 1 : The Formaldehyde results

Test	Result	Test Method
Formaldehyde Content	0.2%	The sample was analysed by UV-Vis Spectrophotometer using Acetylacetone as reagent.

h) Table 2 : The Elemental results

Test	Result	Test Method
Mercury	Not Detected ^a	The sample was digested in Inorganic acid, followed by analysis using Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP-AES)
Lead	Not Detected ^a	
Cadmium	Not Detected ^a	
Chromium	Not Detected ^a	

^a – The method detection limit was 0.01%

i) Table 3 : The Flash Point results

Test	Result	Test Method
Flash point @61°C	No Flash	By Seta Flash

j) Table 4 : The Volatile Organic Compound (VOC) content

Test	Result	Test Method
VOC content ^b	42.5 g/L	By BS EN ISO 11890-2:2006, Paints and varnishes – Determination of content volatile organic compound (VOC) content – Part 2: Gas-chromatographic method.

^b – Volatile organic compound (VOC) means any organic compound having an initial boiling point less than or equal to 250°C measured at a standard pressure of 101,3kPa.

k) Table 5 : The analytical results

Test	Result	Test Method
Halogenated solvent	Not Detected ^c	By Gas Chromatograph – Mass Spectrometry (GC-MS)
Epichlorohydrin	Not Detected ^c	
Aromatic solvent	Not Detected ^d	

^c – The method detection limit was 0.01%

^d – The method detection limit was 0.1%

l) Chemical Resistance of Viscacid BS 3000

Sodium Hydroxide sol. 50%
Potassium Hydroxide sol 10%
Methyl isobutyl ketone 30%
Breake Oil
Hydraulic Oil
Detergent Soul. Conc.
Sugar Solution Saturated
Sodium Chloride sol. 10 %
Sodium Chloride sol. Sat.
Ammonium Chloride sol. 25%
Thomas potash 10%
Epsom salt 10%
Epsom salt saturaterd
Nitro Chalk 10% saturated

m) Slip resistant floor coating system : 'R-12' when tested with DIN 51130

n) Steam and high pressure water cleanable tested to 130°C hot water and 40 bar water pressure

Coloured seal coats (slip retardant)

Working operation	Materials	Tear strength [N/mm ²]		
		Unloaded	Loaded	Slightly loaded
Primer	Viscacid BS 2000 with 5% Acolan ADD 250	2.63 concrete failure	3.93 concrete failure	3.33 concrete failure
Coating	Viscacid BS 3000 with 5% Acolan ADD 250			

o) Flame resistant according to BfL.

B. BASF Mastertop 1705 anti-dust sealer for raised floor
Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect.

- All components of the waterproofing system are non-toxic and do not cause any hazard or injury to operatives or to others in the vicinity of the Site;
- The system is durable coloured coating based on an advanced water-dispersed, twopart epoxy resin. It has been specially formulated to provide a durable coating system.

- The system is suitable for contact with water intended for human consumption. It complies with SS375:1994 and is an approved material by Public Utilities Board of Singapore.
- The system can be applied on a wide variety of substrates such as concrete, masonry, etc. It is recommended as a protective coating on floors and walls where both high levels of aesthetics and chemical resistance are desired. Applications include
 - i. coating areas in contact with potable water;
 - ii. wall coating in pharmaceutical and other areas that must be easy to clean;
 - iii. protection of floors against mild acids, alkalis, mineral oils, etc.;
 - iv. road tunnel wall coatings to improve the light conditions, to facilitate easy maintenance and to protect concrete against chemicals / pollutants;
 - v. tunnel segment protection coating;
 - vi. curing membrane for concrete;
 - vii. light traffic areas in industrial buildings and warehouses.
- The system is odourless and nontoxic, resistant to attack by chemicals, water vapour permeable, nonflammable, color fast and that tools can be cleaned with water.
- The system can be kept for 12 months in original unopened packing when stored indoors. Do not store in direct sunlight and avoid allowing the material to freeze which will render the material unusable.
- The system is available in 10 kg/set.

Components	Supply form	Colour	Density kg/L
MasterTop 1705 PTA	Liquid	Coloured	1.40
MasterTop 1705 PTB	Liquid	Amber	1.01

Mixed Systems	Components	Mix ratio by weight
MasterTop 1705	A:B	1:1
Service Temp	-20°C to 60°C	
Application Temp	Min 10°C	Max 35°C
Pot-life (minutes)	@20°C	@30°C
MasterTop 1705	30	15

Curing Time	@10°C	@20°C	@30°C
MasterTop 1705	7	2	1

Maximum permissible relative humidity	@10°C	>23°C
All components	75%	90%

System	Product	kg/m3	Thickness mm (wft)
1 st coat	MasterTop 1705 Diluted with (approx) 10% water	0.20-0.30	0.15
2 nd and 3 rd coat	MasterTop 1705	0.20-0.25	0.15-0.20

PART 3 EXECUTION

3.01 EXAMINATION/INSPECTION

- A. Applicator shall examine areas and conditions under which special coating work is to be applied and notify Contractor, in writing, of conditions detrimental to proper and timely completion of work. Do not proceed with work until unsatisfactory conditions have been corrected.
- B. Determine alkalinity and moisture content of surfaces to be painted by performing appropriate tests recommended by the special coatings manufacturer. The following is for guidance only and describes a general outline of procedures which shall be followed:
 - 1. If surfaces are found to be sufficiently alkaline to cause blistering and burning of special coating, correct condition before application of special coating.
 - 2. Do not coat over dirt, rust, scale, grease, moisture, scuffed surfaces, and conditions otherwise detrimental to formation of a durable coating film.
 - 3. Do not coat over surfaces where moisture content exceeds that permitted in Manufacturer's printed instructions.
 - 4. Clean floor surfaces scheduled to be coated with a commercial solution of muriatic acid, or other etching cleaner. Flush floor with clean water to neutralize acid, and allow to dry before coating.
- C. Starting Work constitutes acceptance of the existing conditions and the Special Coating Sub-Contractor shall, at his own expense, be responsible for correcting all unsatisfactory and defective Work encountered.
- D. Ensure that all adjacent surfaces, glass/window areas and finish materials are protected and masked from accidental spraying, paint-over or droppings of special coatings.

3.02 PREPARATION

- A. Surfaces to receive the Remmers systems shall be uniformly smooth, clean and dry. Remove any water, dirt or debris and permit to dry. All holes, joints, and cracks in concrete surfaces shall be pointed flush with epoxy mortar and allowed to cure. High spots in concrete or masonry surfaces shall be cut off or ground down. Before application, all dust, dirt and foreign matter, shall be removed and shall be inspected and approved by Remmers' representative, prior to commencing the systems.
- B. Equipment
 - 1. General

All equipment used for the installation of the Systems shall be as recommended by Remmers' manufacturer and as specified herein.
- C. Preparation
 - 1. Condition of Surfaces:

Surfaces to receive the systems shall be free of defects or conditions that will interfere with, or prevent a satisfactory installation of the systems. Concrete surfaces shall be sufficiently dried to receive it.

2. Contaminants

- a. Surfaces shall be thoroughly cleaned and shall be free of water, dew, frost, dust, dirt, oil, grease, laitance and other foreign matter. After the cleaning done by Contractor is not satisfactory, then the Contractor shall clean concrete substrate by grinding to remove laitance, concrete curing compounds, and other contaminants.
- b. Dirt and dust shall be removed with an air compressor or industrial type vacuum cleaner of methods approved by Remmers' manufacturer.

3. Repairs:

The Contractor before application of the systems shall repair voids, cracks, holes and other damaged surfaces with epoxy repair mortar.

3.03 INSTALLATION/APPLICATION

A. REMMERS Viscacid BS2000 Primer and Silica Sand and REMMERS Viscacid BS3000 top coat for pump/plant room

1. Surface Finish Requirement

- a. The base concrete floor shall be of minimum Grade 30 strength, well compacted, vibrated and screeded to the level and finishing as specify by the Architect.
- b. The concrete must allow for curing of minimum 28 days.
- c. The entire floor surface must be cleaned and free from cement laitance, oil, grease and any of the systems.

2. Preparation Requirement

- a. Prior to the application of Remmers Systems, the entire floor surface to be prepared by light vacuum shot blasting and/or concrete grinding to receive the Systems.

3. Application Procedure

- a. Apply Floor Hardener. Details to be referred to Section 2.1 Cast Decks and Underlayment.
- b. Remove any excess sand and apply two coats of Viscacid BS 2000 (Epoxy BS2000) to the entire area as an intermediate coat. Allow to dry for min. 8 hours between coats.
- c. Apply one coat of Viscacid BS 3000 (Epoxy BS3000) as a finishing seal coat. Allow to cure.

3.04 CLEANING

- A. As Work proceeds, and upon completion, promptly remove coating where spilled, splashed, or spattered. During progress of Work keep premises free from any unnecessary accumulation of tools, equipment, surplus material, and debris.
- B. Upon completion of Work leave premises neat and clean.
- C. Legally transport and dispose of all coatings, solvents and other coating productions in accordance with local Environment Protection regulations.

3.05 PROTECTION

- A. Protect other surfaces from paint and damage. Repair damage as a result of inadequate or unsuitable protection.
- B. Install "Wet Paint. Do not touch." signs in English and Chinese Languages and installs barrier tapes around painted areas and items.
- C. Following application of Nippon weatherbond paint system, the Contractor shall provide adequate protection to the coating against abrasive contact for not less than seven days, and against rain for the first 48 hours after application.

END OF SECTION